Unit-4 - Random Processes and Stationary Processes	**12 Hour**
Random Processes-Introduction, Classification of random processes, Distribution of the process, Averages of the process, Stationary, SSS, WSS processes, Problems on stationary and SSS processes, Problem, Problems on WSS process, Problems on WSS process, Autocorrelation function -properties, Proof of properties, Problems on autocorrelation function, Application of autocorrelation function, Cross-correlation-properties, Proof of properties, Problems on Cross-correlation function, Ergodicity, Mean ergodic process, Mean ergodic theorem, Applications of random process in engineering.	
Unit-5 - Spectral Density of Random Process and Linear System with Random Inputs	**12 Hour**
Power spectral density function- properties, Proof of properties, Problems on power spectral density function, Problems on power spectral density function, Power density spectrum, Problems based on power density spectrum, Linear systems with random inputs, Representation of system in the form of convolution, Unit impulse response of the system, Properties, Applications of unit impulse function, Einstein Weiner-Khinchine Relationship, Cross-power density spectrum-problems, Cross-power density spectrum Cross-power density spectrum, Applications of power spectral density functions in engineering, Applications of power spectral density functions in engineering.	

Learning Resources	1. *A. Papoulis, S. Unnikrishna Pillai, Probability, Random Variables and Stochastic Processes 4th Edition, Mcgraw Hill, 2002.* 2. *Henry Stark, Probability and Random Processes with Applications to Signal Processing, Third Edition, Pearson 2004* 3. *Sheldon Ross, A first course in Probability, Sixth Edition, 2011.*	4. *S.C. Gupta, V.K. Kapoor, Fundamentals of Mathematical Statistics, Sultan Chand & Sons, 11th Edition, 2015.* 5. *Jay L DeVore, Probability and Statistics for Engineering and the Sciences, 8th Edition, Cengage Learning India Pvt. Ltd, 2012.* 6. *T. Veerarajan, Probability, Statistics and Random Processes with Queueing Theory and Queueing Networks, 4th Edition, McGraw-Hill Education, New Delhi, 2015.*

Learning Assessment

	Bloom's Level of Thinking	*Continuous Learning Assessment (CLA)*				*Summative Final Examination (40% weightage)*	
		Formative CLA-1 Average of unit test (50%)		*Life-Long Learning CLA-2 (10%)*			
		Theory	*Practice*	*Theory*	*Practice*	*Theory*	*Practice*
Level 1	*Remember*	*20%*	-	*20%*	-	*20%*	-
Level 2	*Understand*	*20%*	-	*20%*	-	*20%*	-
Level 3	*Apply*	*30%*	-	*30%*	-	*30%*	-
Level 4	*Analyze*	*30%*	-	*30%*	-	*30%*	-
Level 5	*Evaluate*	-	-	-	-	-	-
Level 6	*Create*	-	-	-	-	-	-
	Total	*100 %*		*100 %*		*100 %*	

Course Designers

Experts from Industry	**Experts from Higher Technical Institutions**	**Internal Experts**
1. *Mr. Madhan Shanmugasundaram, Infosys Technologies, madshan@gmail.com*	1. *Prof. Y.V.S.S. Sanyasiraju, IIT Madras, sryedida@iitm.ac.in*	1. *Dr. V. Subburayan, SRMIST*
	2. *Prof. K.C. Sivakumar, IIT Madras, kcskumar@iitm.ac.in*	2. *Dr. G. Vijayalakshmi, SRMIST*

Course Code	21MAB204T	Course Name	PROBABILITY AND QUEUEING THEORY	Course Category	B	BASIC SCIENCES	L	T	P	C
							3	1	0	4

Pre-requisite Courses	*Nil*	Co- requisite Courses	*Nil*	Progressive Courses	*Nil*
Course Offering Department	*Mathematics*	Data Book / Codes / Standards		*Nil*	

Course Learning Rationale (CLR):	*The purpose of learning this course is to:*
CLR-1:	*describe the properties of a random variable*
CLR-2:	*gain knowledge of discrete and continuous distributions*
CLR-3:	*understand the basic concepts of two-dimensional random variables, correlation, and regression lines*
CLR-4:	*interpret the system characteristics of queueing models*
CLR-5:	*create Markov chains and investigate stationary state distribution*

Course Outcomes (CO):	*At the end of this course, learners will be able to:*	Program Outcomes (PO)												Program Specific Outcomes		
		1	2	3	4	5	6	7	8	9	10	11	12			
		Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning	PSO-1	PSO-2	PSO-3
CO-1:	*evaluate the characteristics of discrete and continuous random variables and apply them in science and engineering*	*3*	*3*	-	-	-	-	-	-	-	-	-	-	-	-	-
CO-2:	*identify the random variables and model them using various distributions*	*3*	*3*	-	-	-	-	-	-	-	-	-	-	-	-	-
CO-3:	*infer results from two dimensional random variables which describe real life phenomenon*	*3*	*3*	-	-	-	-	-	-	-	-	-	-	-	-	-
CO-4:	*examine the significant results of various queueing models*	*3*	*3*	-	-	-	-	-	-	-	-	-	-	-	-	-
CO-5:	*determine the transition probabilities and classify the states of the Markov chain*	*3*	*3*	-	-	-	-	-	-	-	-	-	-	-	-	-

Unit-1 - Random Variables	***12 Hour***
Probability concepts - Discrete and continuous random variables - Probability distribution function, Cumulative distribution function - Moments - Central and raw moments, Expectation and variance - Moment generating function (MGF) –Tchebycheff's inequality – Function of a random variable.	
Unit-2 - Theoretical Distributions	***12 Hour***
Discrete distribution - Binomial distribution, Poisson distribution - MGF, Mean, Variance, Theoretical frequencies and applications – Continuous distribution - Exponential and normal distributions – MGF, Mean, Variance and applications.	
Unit-3 - Two-Dimensional Random Variables	***12 Hour***
Joint distributions - Marginal and conditional distributions – Covariance – Correlation and linear regression – Central limit theorem (for independent and identically distributed random variables)	
Unit-4 - Queueing Theory	***12 Hour***
Queueing theory – Characteristics of a queueing Model – Kendal's notation – Poisson queues - (M/M/1): (∞/FIFO) Model – System characteristics – Applications - (M/M/s): (∞/FIFO) Model - System characteristics – Applications - (M/M/1): (k/FIFO) Model - System characteristics – Applications.	
Unit-5 - Markov Chain	***12 Hour***
Markov process – Markov chain – One step transition probability matrix – Chapman Kolmogorov theorem – Limiting probabilities – Classification of states of a Markov chain.	